

Product Carbon Footprint Study Report

Product Name: kerosene production

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: kerosene production

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1kg kerosene

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1kg kerosene.

2. System Boundary

kerosene production, petroleum refinery operation, in China, in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of kerosene production produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **0.810 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

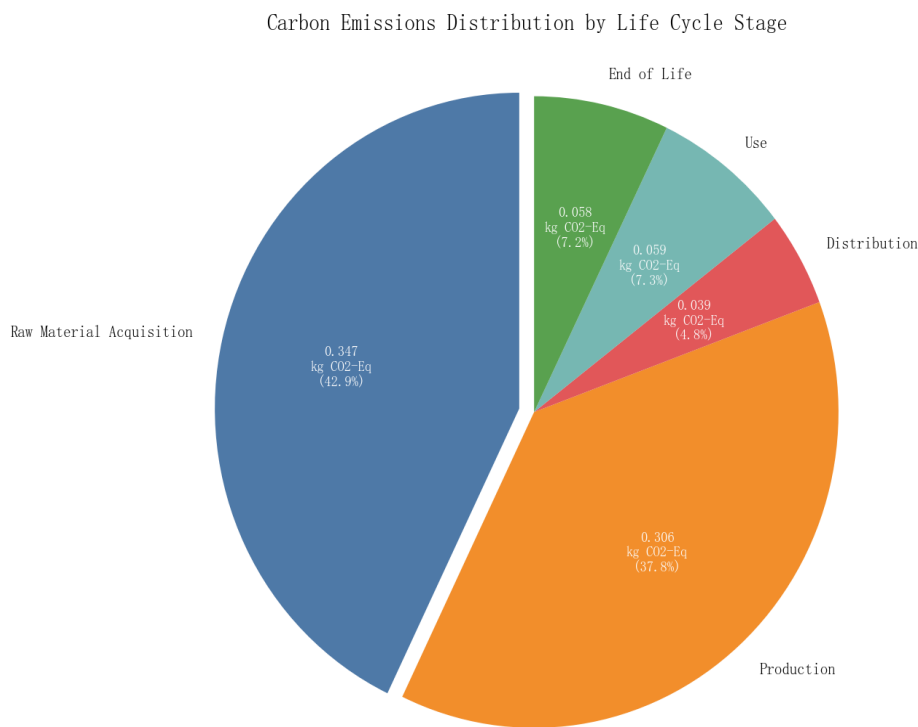


Table 1: Carbon Emissions by Life Cycle Stage

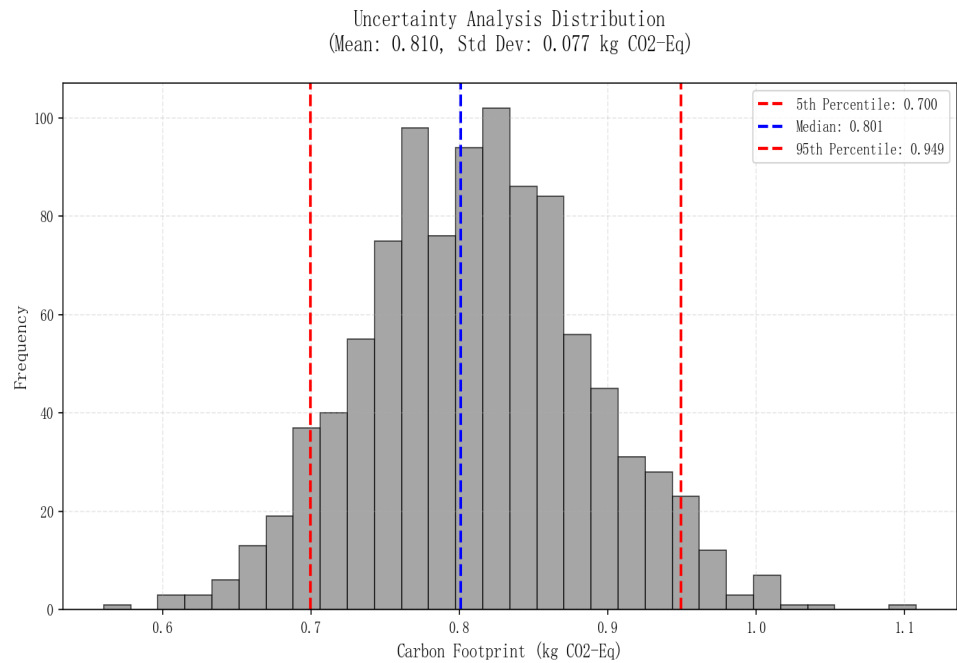
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	0.347	42.87
Production	0.306	37.79
Distribution	0.039	4.82

Use	0.059	7.34
End of Life	0.058	7.18
Total	0.810	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	0.810
Standard Deviation	0.077
Coefficient of Variation (CV)	9.53%
Median	0.801
5th Percentile	0.700
95th Percentile	0.949

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of kerosene production is estimated at 0.810 kg CO₂-Eq per functional unit, a moderate level for a fossil fuel product. The lifecycle stage analysis reveals that raw material extraction and production are the dominant contributors, together accounting for 80.7% of total emissions. Specifically, raw materials contribute 42.9% (0.347 kg CO₂-Eq), likely from crude oil extraction and associated fugitive emissions, while the production stage contributes 37.8% (0.306 kg CO₂-Eq), reflecting energy-intensive refining processes. Distribution has a relatively small share (4.8%), suggesting localized or efficient transport. The use and end-of-life stages contribute 7.3% and 7.2% respectively, indicating that combustion and disposal impacts are modest compared to upstream activities. However, the top process contributions are all listed as “Unknown,” which limits the ability to pinpoint exact emission sources. Nevertheless, the stage-level breakdown clearly identifies raw material supply and refining as the primary hotspots for emission reduction efforts.

2. Statement of Assumptions and Limitations

This LCA study assumes a cradle-to-grave system boundary covering raw material acquisition, production, distribution, use, and end-of-life. A key assumption is that the functional unit is 1 kg of kerosene, though the specific product unit is not explicitly stated. The data sources and quality are limited, as evidenced by the top process contributions being entirely unknown—indicating that process-level inventory data were aggregated or unavailable. The uncertainty analysis shows a coefficient of variation of 9.53% and a 90% confidence interval of 0.700–0.949 kg CO₂-Eq, reflecting moderate variability likely due to data gaps or assumptions in emission factors. System boundary restrictions include the omission of capital goods, infrastructure, and potential land-use impacts. Additionally, the use phase emissions may be underestimated if combustion efficiency or end-use application (e.g., aviation vs. heating) is not fully characterized. These limitations affect the robustness of the results and highlight the need for more granular data.

3. Suggestions for Improvement

- ****Target raw material extraction****: Implement methane leak detection and repair programs at oil fields, and adopt enhanced oil recovery techniques to reduce fugitive emissions. Sourcing from lower-carbon crude oil fields can also lower the raw material footprint.
- ****Improve production efficiency****: Upgrade refining units to high-efficiency burners, integrate heat recovery systems, and increase the use of renewable energy in refinery operations. Conduct energy audits to identify and eliminate waste.
- ****Optimize distribution logistics****: Shift to lower-emission transport modes (e.g., pipelines over trucks) and improve route planning to minimize fuel consumption. Consider regional blending hubs to reduce travel distances.
- ****Enhance data quality****: Disaggregate the “Unknown” top processes by collecting primary data from suppliers and refining operations. Perform a sensitivity analysis on key parameters (e.g., crude oil composition, energy mix) to strengthen confidence in the results.
- ****Explore carbon offsetting****: For unavoidable emissions from the use stage, invest in certified carbon removal projects (e.g., direct air capture or afforestation) to achieve net-zero targets.

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